ML SOCIETY · KAGGLE SESSIONS

LightGBM vs XGBoost

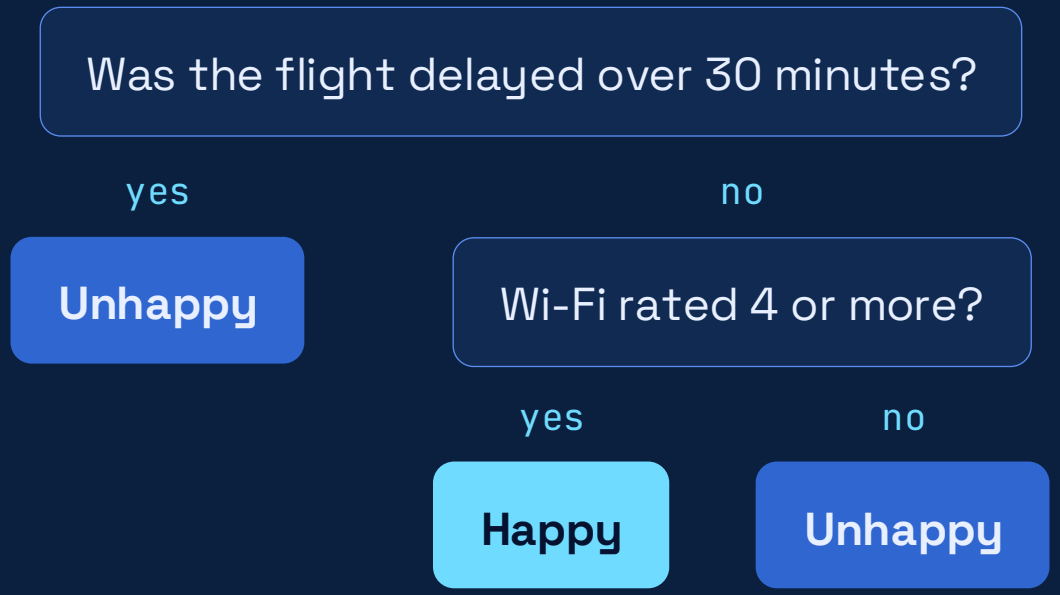
Two ways to grow a boosted tree, and why our first submission used LightGBM.

LightGBM · what we used

XGBoost · today

FIRST, THE BUILDING BLOCK

What is a decision tree?



A tree is a chain of yes/no questions. Each answer sends a row down a branch until it reaches a prediction. The questions/thresholds at every level are learned, to optimally split samples.

THE SHARED IDEA

Both are gradient boosting

Tree 1

Makes a rough first guess.

→

Tree 2

Fixes the mistakes of tree 1.

→

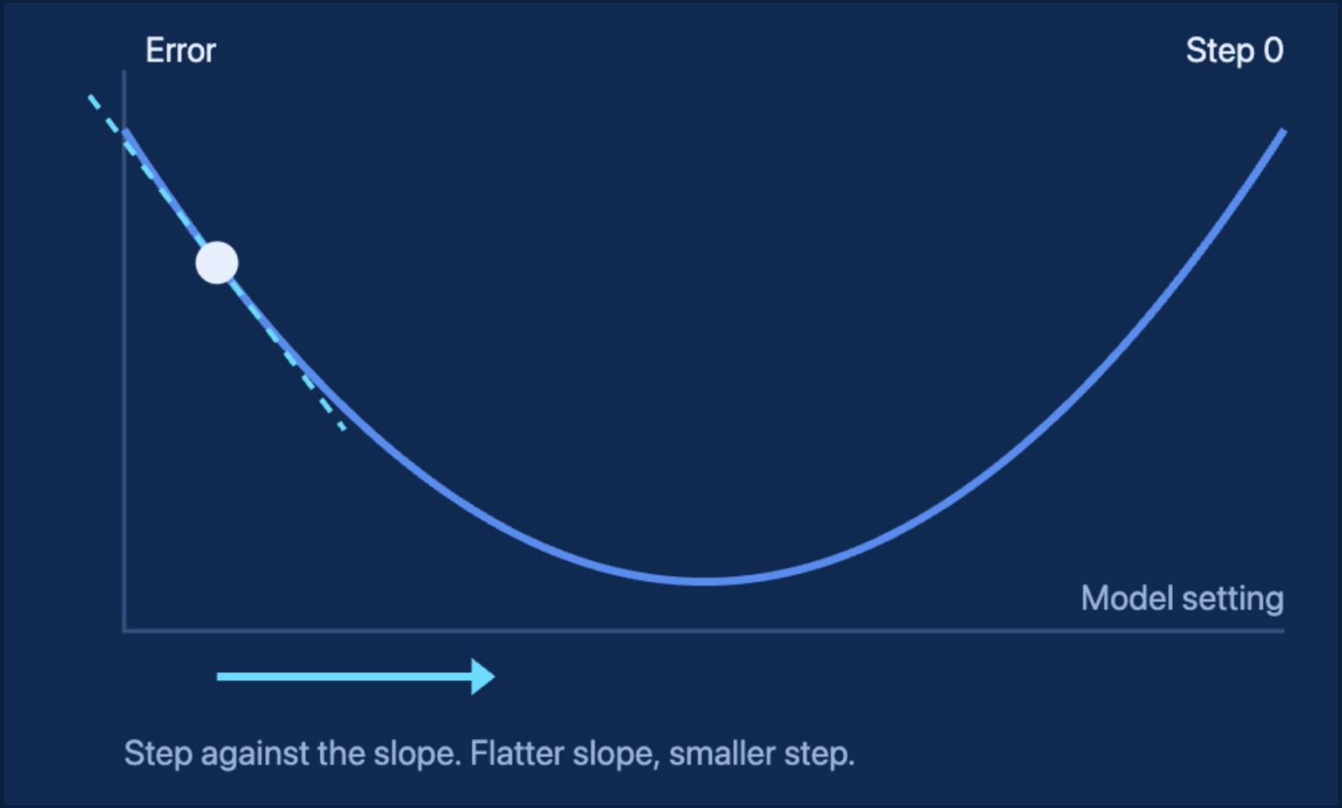
Tree 3, 4, 5...

Each one fixes what is still wrong.

Hundreds of small, weak trees add up to a strong model. Like a study group where each person only reviews the questions the last one got wrong.

WHAT "GRADIENT" MEANS

Walking downhill to less error



Loss

How wrong the model is. Lower is better.

Gradient

The slope: which way is uphill. We step the opposite way.

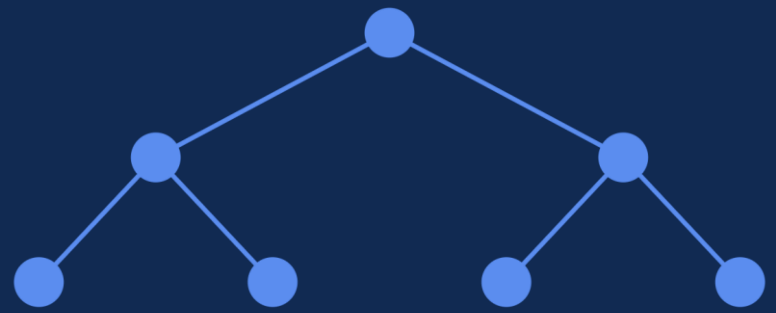
Boosting

Each new tree is one step downhill.

THE BIG DIFFERENCE

How the tree grows

XGBoost: level by level



Splits every node on a level before going deeper. Balanced and safe.

LightGBM: leaf by leaf



Always splits the one leaf that helps most. Goes deeper only where it matters.

WHY LIGHTGBM IS FAST

Fewer choices to check

Histograms

Instead of testing every exact value as a split point, it sorts values into about 255 buckets and tests only the bucket edges.

Leaf-wise growth

Splits go where the error is largest, so fewer splits are needed for the same improvement.

Small print

XGBoost has caught up. Its histogram mode is the default since version 2.0, so the gap is smaller today.

SIDE BY SIDE

What changes for you

	LightGBM	XGBoost
Tree growth	Leaf-wise	Level-wise (default)
Big tables	Usually faster	Fast, close in recent versions
Small datasets	Overfits more easily	Safer out of the box
Category columns	Built in	Opt in: enable_categorical=True

OVERFITTING AGAIN

One knob controls complexity

LightGBM

```
num_leaves=31
```

Maximum number of leaves in each tree.

XGBoost

```
max_depth=6
```

Maximum number of levels in each tree.

Bigger numbers mean a more complex model: more room to memorise, just like last session. Leaf-wise trees overfit faster, so keep an eye on this one in LightGBM.

TRY IT

Same data, different model

```
from lightgbm import LGBMClassifier
from xgboost import XGBClassifier

model = LGBMClassifier() # what we used
model = XGBClassifier() # one-line swap
model.fit(X_train, y_train)
```

Both work like any scikit-learn model. Score each on the same validation split.

TAKEAWAY

Neither wins every time. Your validation score decides.

LightGBM: fast, a great first model. XGBoost: steady, a strong second opinion.

ACTION NEEDED · 10 DAYS LEFT

Everyone: check your team name and your submissions. We can’t see every name on the leaderboard.

IT'S YOUR TURN!

Get set up again

1

Pull the repo

Get the new notebook folder.

```
git pull
```

2

Copy the files again

From the git folder folder into your local working folder, like the last two weeks.

Create a NEW local week-03 folder!

3

Rebuild your environment

Delete the old virtual environment, make a new one, reinstall.

```
python -m venv .venv
source .venv/bin/activate      MacOS/Linux
.venv/scripts/activate         Windows
pip install -r requirements.txt
```